

MATH 310, Section 3.4 Solutions

2. $A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

(a) $A\mathbf{x}_1 = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 5\mathbf{x}_1$

Thus $\lambda_1 = 5$ is an eigenvalue with corresponding eigenvalue $\mathbf{x}_1$.

(b) $A\mathbf{x}_2 = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 1\mathbf{x}_2$

Thus $\lambda_2 = 1$ is an eigenvalue with corresponding eigenvalue $\mathbf{x}_2$.

12. $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

(a) The characteristic polynomial is $|\lambda I - A| = \begin{vmatrix} \lambda - 2 & 0 & -1 \\ 0 & \lambda - 3 & -4 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = \boxed{(\lambda - 2)(\lambda - 3)(\lambda - 1)}$

(b) The eigenvalues are the roots of the characteristic polynomial, and are thus $\boxed{\lambda_1 = 1, \lambda_2 = 2, \text{ and } \lambda_3 = 3.}$

(c) i. Eigenvectors for $\lambda = 1$:

Need to solve $(1I - A)\mathbf{x} = \mathbf{0}$

$$\begin{bmatrix} -1 & 0 & -1 & 0 \\ 0 & -2 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The solutions are } x = -z, y = -2z, z = t$$

$$\boxed{\text{so the eigenvectors for } \lambda = 1 \text{ are } \begin{bmatrix} -t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}}$$

ii. Eigenvectors for $\lambda = 2$:

Need to solve $(2I - A)\mathbf{x} = \mathbf{0}$

$$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The solutions are } x = t, y = 0, z = 0$$

$$\boxed{\text{so the eigenvectors corresponding to } \lambda = 2 \text{ are } \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

iii. Eigenvectors for $\lambda = 3$:

Need to solve $(3I - A)\mathbf{x} = \mathbf{0}$

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The solutions are } x = 0, y = t, z = 0$$

$$\boxed{\text{so the eigenvectors corresponding to } \lambda = 3 \text{ are } \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$